Yitong Zhang (张奕彤)

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https://zhangyitonggg.github.io/



Education

- 2026.09 - 2031.06 (expected), Ph.D. Student, College of AI, Tsinghua University, under the supervision of Prof. Jia Li
- 2022.09 - 2026.06, Undergraduate Student, School of Computer Science and Engineering, Beihang University, under the supervision of Prof. Xianglong Liu and Prof. Aishan Liu

Research Interests

I aim to build more capable and trustworthy intelligent software systems, with a particular focus on LLM-based agents. ❶ On the capability side, I study how these systems can reason, use tools, and carry out complex, long-horizon tasks. ❷ On the trustworthiness side, I investigate how to ensure that their actions remain aligned with user intent, robust to adversarial interference, and subject to verifiable constraints as their capabilities grow.

Honors and Awards

- **CAST Young Science and Technology Talent Cultivation Program (Doctoral Student Special Plan) (中国科协青年科技人才培育工程博士生专项计划)**, 2026
- **Tsinghua Doctoral Fellowship for Future Scholar (清华大学未来学者奖学金; Only 4 recipients in the College of AI)**, 2026
- **Shen Yuan Medal (沈元奖章; Top 0.1%; Only 10 undergraduates university-wide; The highest scholarship of Beihang University)**, 2025
- **Beihang Youth May Fourth Medal (北航青年五四奖章)**, 2026
- **China National Scholarship (three-time recipient)**, 2023, 2024, 2025
- **CCF Elite Collegiate Award**, 2026
- **Outstanding Student of Beijing**, 2025

Grants

2026.09 | Large Model Driven End-to-End Autonomous Driving Model Adversarial Attacks
2024.10 | - **Principal Investigator**, Beijing Natural Science Foundation (Grant No. QY24136).
- One of 40 students selected university-wide.

Selected Publications

**** indicates author with equal contribution.***

- Environmental Injection Attacks against GUI Agents in Realistic Dynamic Environments
  Yitong Zhang, Ximo Li, Liyi Cai, Jia Li
  ISSTA 2026
- DAVSP: Safety Alignment for Large Vision-Language Models via Deep Aligned Visual Safety Prompt
  Yitong Zhang, Jia Li, Liyi Cai, Ge Li
  AAAI 2026 (Oral), Top 4%
- Lookahead-then-Verify: Reliable Constrained Decoding for Diffusion LLMs under Context-Free Grammars
  Yitong Zhang, Yongmin Li, Yuetong Liu, Jia Li, Xiaoran Jia, Zherui Li, Ge Li
  ISSTA 2026
- Grammar-Constrained Decoding Can Jailbreak LLMs into Generating Malicious Code
  Yitong Zhang, Shiteng Lu, Jia Li
  ICSE 2027

- Beyond Autoregression: An Empirical Study of Diffusion Large Language Models for Code Generation
  Chengze Li*, **Yitong Zhang* (Equal Contribution)**, Jia Li, Liyi Cai, Ge Li
  TOSEM
- DiffuCD: Efficient Contrastive Decoding for Diffusion Language Models
  Jia Li, **Yitong Zhang (Student First Author)**, Zhi Jin
  Journal of Software
- To See is Not to Master: Teaching LLMs to Use Private Libraries for Code Generation
  Yitong Zhang, Chengze Li, Ruize Chen, Guowei Yang, Xiaoran Jia, Yijie Ren, Jia Li
  TOSEM Major Revision
- Safe Actions Alone Do Not Ensure Safe Agents: Identifying Unfulfilled Obligations with Guard Models
  Youwei Feng, **Yitong Zhang (Project Leader)**, Yuetong Liu, Jia Li
  FSE 2027 Under Review
- AI-Driven Self-Evolving Software: A Promising Path Toward Software Automation
  Liyi Cai*, Yijie Ren*, **Yitong Zhang* (Equal Contribution)**, Jia Li
  Preprint
- *Additional publications can be found on my **Google Scholar profile**.*

Teaching and Service

- Reviewer: The IEEE Transactions on Information Forensics and Security
- Reviewer: The 40th Annual AAAI Conference on Artificial Intelligence
- Teaching Assistant: LLM-Driven Software Development, Tsinghua University, Fall 2026, Jia Li
- Teaching Assistant: Discrete Mathematics, Beihang University, Spring 2025, Lei Shi
- Teaching Assistant: Computer Organization, Beihang University, Fall 2024, Han Wan
- Senior Teaching Assistant: Data Structures and Programming, Beihang University, Spring 2024, Ying Li